

Control Systems A

Week 2 - Mathematical Modelling of Physical Systems

UoN - FEE 3411

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What we are doing today

- **Hour 1 — the framework, and electrical networks.** What a differential-equation model looks like in general, and how Kirchhoff's laws produce one for any network.
- **Hour 2 — mechanical systems, gears, and the servomotor.** The same framework applied to translational and rotational systems, then combined — electrical and mechanical together — into the d.c. servomotor promised at the end of Week 1. Linearization closes the lecture.

Why this matters

Every method from Week 7 onward is applied to a model, never to the physical system. This week is where that model comes from.

Syllabus: *Inputs, plant and outputs. . . Mathematical models. The characteristic equation. Linear approximation to nonlinear systems. Analysis of linear (causal) systems. . .* ELO 1, ELO 2.

The modelling framework

Every model has the same shape

Whatever the physics, a single-input–single-output plant model is

$$a_n y^{(n)} + \cdots + a_1 \dot{y} + a_0 y = b_m u^{(m)} + \cdots + b_1 \dot{u} + b_0 u,$$

a linear, constant-coefficient ODE.

- **Linear** — superposition holds; no y^2 , no $u\dot{y}$, no $\sin y$.
- **Time-invariant** — the coefficients don't depend on t .
- **Causal** — the output at time t depends only on the input up to time t .

Khalil, p. 10.

The characteristic equation

Key idea

Set the input to zero and replace each $y^{(k)}$ by s^k :

$$a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0 = 0.$$

It depends only on the **plant** — never on the input. Its roots govern every natural response the system can produce, which is why Weeks 7–12 are all, in one way or another, about the location of these roots.

Two systems, one characteristic equation

Common pitfall

Two different physical systems with the *same* characteristic equation behave **identically** in their dynamics — same natural frequency, same damping, same transient shape — even though the physical quantities, units and steady-state gain can be completely different.

You will meet exactly this pairing later this lecture, between an electrical network and a mechanical system. It is the reason the same mathematics serves electrical, mechanical, thermal and fluid systems alike.

Three routes to a model (Khalil, p. 6): **first principles** (this week), **identification** from measured data, or **both**.

Modelling electrical networks

Kirchhoff's laws as the starting point

KVL: the signed sum of voltages round any closed loop is zero.

KCL: the signed sum of currents into any node is zero.

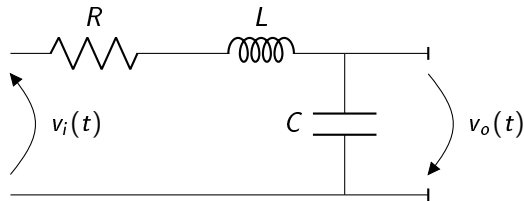
Element relations, passive sign convention:

$$v_R = Ri, \quad v_L = L \frac{di}{dt}, \quad i_C = C \frac{dv_C}{dt}.$$

Key idea

Writing KVL (or KCL) with these relations substituted in is the differential-equation model. No new physics is needed anywhere in this unit for an electrical network — only bookkeeping.

Worked example: the series RLC network



Cf. Khalil Fig. 2-4, p. 17, relabelled with output v_o across C .

Series RLC — KVL, then eliminate

Same current $i(t)$ through all three elements:

$$v_i = Ri + L \frac{di}{dt} + v_o, \quad i = C \frac{dv_o}{dt}.$$

Eliminate i :

Worked example — series RLC

$$\boxed{LC \ddot{v}_o + RC \dot{v}_o + v_o = v_i} \quad \Rightarrow \quad \frac{V_o(s)}{V_i(s)} = \frac{1}{LCs^2 + RCs + 1}$$

For $R = 6 \Omega$, $L = 2 \text{ H}$, $C = 0.05 \text{ F}$: $s^2 + 3s + 10 = 0$, roots $s = -1.5 \pm j2.78$.

Common pitfall

Decide *first* which quantity is the output and eliminate everything else in its favour. A mixed equation in both i and v_o cannot be read off against the general form.

A preview of next week

The method generalises with no dead ends: KVL/KCL per loop or node, including networks with op-amps, using the ideal op-amp conditions in place of a KCL equation.

Key idea

Next week: replace each element by its **impedance** (R , Ls , $1/Cs$) and apply Kirchhoff's laws directly in the s -domain — reaching the same transfer function without ever writing a differential equation. This week the differential equation is the point, because the characteristic equation is defined from it directly.

Translational mechanical systems

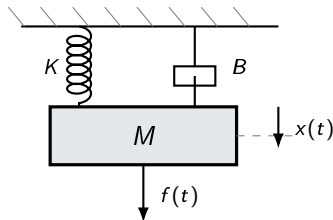
Spring, damper, mass

$$f_{\text{spring}} = Kx, \quad f_{\text{damper}} = B\dot{x}, \quad f_{\text{net}} = M\ddot{x}.$$

A **free-body diagram** — every force on the mass, with its correct sign — turns this into an equation of motion via Newton's second law.

Khalil, p. 24.

Worked example: mass-spring-damper



Newton's second law

$$M\ddot{x} + B\dot{x} + Kx = f$$

$$\frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Bs + K}$$

For $M = 2 \text{ kg}$, $B = 6 \text{ N s m}^{-1}$, $K = 18 \text{ N m}^{-1}$: $s^2 + 3s + 9 = 0$, roots $s = -1.5 \pm j2.60$.

The force–voltage analogy

Writing the RLC equation in charge, $i = \dot{q}$:

$$L\ddot{q} + R\dot{q} + \frac{1}{C}q = v_i \quad \longleftrightarrow \quad M\ddot{x} + B\dot{x} + Kx = f.$$

Key idea

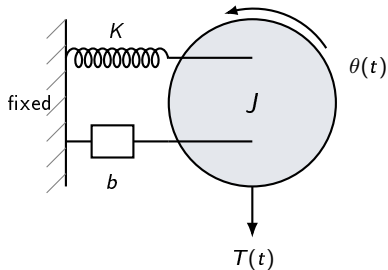
The *same* equation, under $M \leftrightarrow L$, $B \leftrightarrow R$, $K \leftrightarrow 1/C$. This is why one body of mathematics — $s^2 + 2\zeta\omega_n s + \omega_n^2$, dominating Weeks 5 onward — serves electrical and mechanical engineers alike.

It says nothing about the **gain**: two analogous systems can have wildly different d.c. gains and units. You will use this comparison directly in Assignment 1.

Rotational mechanical systems

Torque, angle, inertia

$$T_{\text{spring}} = K\theta, \quad T_{\text{damper}} = b\dot{\theta}, \quad T_{\text{net}} = J\ddot{\theta}.$$



Khalil, p. 53; Nise §2.6, p. 69.

Worked example: rotational system

Newton's second law for rotation

$$\boxed{J\ddot{\theta} + b\dot{\theta} + K\theta = T} \qquad \frac{\theta(s)}{T(s)} = \frac{1}{Js^2 + bs + K}$$

Key idea

With no restraining spring ($K = 0$) — a motor shaft driving a free inertia — this reduces to

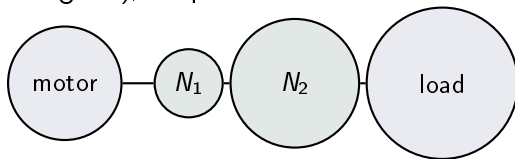
$$\frac{\theta(s)}{T(s)} = \frac{1}{s(Js + b)} :$$

an integrator in series with a first-order lag, the shape flagged for every positional actuator since Week 1.

Gear trains

Why gears: torque for speed

Two meshed gears, $N_1\theta_1 = N_2\theta_2$. With $n = \theta_2/\theta_1 = N_1/N_2$: speeds are in ratio n ; since power is conserved (ideal gears), torques are in the *inverse* ratio $T_2/T_1 = 1/n$.



Khalil, p. 35; Nise Fig. 2.27, p. 74.

Reflecting inertia through the gears

Motor (J_m, b_m) drives, through ratio $n = \theta_L/\theta_m$, a load (J_L, b_L) . Eliminating the load variables:

$$(J_m + n^2 J_L)\ddot{\theta}_m + (b_m + n^2 b_L)\dot{\theta}_m = T_m$$

$$J_{\text{eq}} = J_m + n^2 J_L, \quad b_{\text{eq}} = b_m + n^2 b_L$$

Key idea

Load inertia and friction are seen at the motor shaft scaled by n^2 , not by n . At $n = 1/10$ a load one hundred times the motor's own inertia contributes only as much again.

Two ways to get gears wrong

Common pitfall

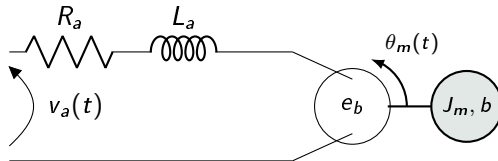
- Reflecting inertia by n instead of n^2 .
- Inverting the ratio — using θ_m/θ_L where $n = \theta_L/\theta_m$ is wanted.

Both are easy to catch: a reduction gear ($n < 1$) should make the equivalent inertia only *slightly* larger than the motor's own, never larger than the load's, and should never produce an implausibly large time constant.

The armature-controlled d.c. servomotor

The servomotor, at last

The system promised at the end of Week 1: electrical modelling (§ Electrical networks) combined with rotational modelling (§ Rotational systems), coupled by two electromechanical constants.



Cf. Khalil Fig. 2-29, p. 37, and Nise Fig. 2.35, p. 79.

The two coupled equations

Electrical (KVL round the armature circuit):

$$v_a = R_a i_a + L_a \frac{di_a}{dt} + e_b, \quad e_b = K_b \dot{\theta}_m.$$

Mechanical (Newton's second law for rotation):

$$T_m = K_t i_a = J_m \ddot{\theta}_m + b \dot{\theta}_m.$$

Why this model is linear

(i) field flux constant $\Rightarrow T_m = K_t i_a$, no product of two varying quantities; (ii) friction purely viscous — Coulomb friction and stiction neglected.

Eliminating the current

Laplace transform, eliminate $I_a(s)$:

The full transfer function

$$\frac{\theta_m(s)}{V_a(s)} = \frac{K_t}{s[(R_a + L_a s)(J_m s + b) + K_t K_b]}$$

The free s outside the bracket means the motor is an **integrator**: a constant armature voltage gives a constant *speed*. Exactly the $K/[s(\tau s + 1)]$ shape from Week 1.

The reduced form: $L_a \rightarrow 0$

Negligible armature inductance

$$\boxed{\frac{\theta_m(s)}{V_a(s)} = \frac{K_m}{s(\tau_m s + 1)}}, \quad K_m = \frac{K_t}{R_a b + K_t K_b}, \quad \tau_m = \frac{J_m R_a}{R_a b + K_t K_b}$$

Common pitfall

Khalil's textbook form also sets $b \rightarrow 0$: $K_m = 1/K_b$, $\tau_m = J_m R_a / (K_t K_b)$. **That extra step is not taken here, nor in Assignment 1** — the motor's own friction b is kept. Check which simplification a question wants before quoting a formula from memory.

Worked example: servomotor constants

For $R_a = 5 \Omega$, $K_t = 0.6 \text{ N m A}^{-1}$, $K_b = 0.6 \text{ V s rad}^{-1}$, $J_m = 0.01 \text{ kg m}^2$, $b = 0.02 \text{ N m s rad}^{-1}$:

$$R_a b + K_t K_b = (5)(0.02) + (0.6)(0.6) = 0.46,$$

$$K_m = \frac{0.6}{0.46} = 1.30 \text{ rad V}^{-1} \text{ s}^{-1}, \quad \tau_m = \frac{(0.01)(5)}{0.46} = 0.109 \text{ s}.$$

Key idea

Most of the denominator comes from $K_t K_b$, not $R_a b$: the motor's own back e.m.f. supplies most of its damping — the electromechanical version of the a.c. servomotor point from Week 1.

Driving a load through a gear train

Combine gearing with the servomotor: load J_L through ratio $n = \theta_L/\theta_m$, load friction negligible
 $\Rightarrow$ every J_m in the reduced formula is replaced by

$$J_{\text{eq}} = J_m + n^2 J_L.$$

Worked example — geared servomotor

Same motor, load $J_L = 2 \text{ kg m}^2$, $n = 1/5$:

$$J_{\text{eq}} = 0.01 + \left(\frac{1}{5}\right)^2 (2) = 0.09 \text{ kg m}^2, \quad \tau'_m = \frac{(0.09)(5)}{0.46} = 0.978 \text{ s}.$$

Roughly nine times slower than unloaded — and without the gearing ($n = 1$) τ_m would exceed 21 s: far too sluggish for a servo.

Linear approximation to nonlinear systems

Why linearize

Real components are rarely linear: a spring stiffens, a valve's flow is not proportional to opening, a motor saturates. Yet almost every method in this unit assumes the linear ODE from the start of the lecture.

Key idea

Near an operating (equilibrium) point, a smooth nonlinear function can be replaced by its tangent: a first-order Taylor expansion. Good *only* for small excursions, and a linearization about one operating point does not, in general, apply near a different one.

Khalil, p. 50; Nise §2.11, p. 89.

Static and dynamic linearization

Static, $y = f(x)$ about (x_0, y_0) :

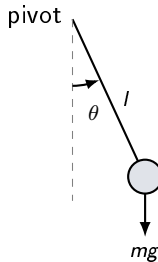
$$\Delta y \approx f'(x_0) \Delta x.$$

This is exactly the a.c. servomotor torque–speed expansion from Week 1, $\Delta T_m = K\Delta E - f\Delta\dot{\theta}$, done without naming the method.

Dynamic, $\dot{x} = f(x, u)$ about equilibrium (x_0, u_0) with $f(x_0, u_0) = 0$:

$$\Delta \dot{x} \approx \left. \frac{\partial f}{\partial x} \right|_0 \Delta x + \left. \frac{\partial f}{\partial u} \right|_0 \Delta u.$$

Worked example: linearizing the pendulum



$$ml\ddot{\theta} = -mg \sin \theta - k l \dot{\theta}, \text{ equilibria at } \sin \theta = 0: \theta = 0 \text{ and } \theta = \pi.$$

Cf. Khalil Fig. 2-44, p. 51.

Same-looking equations, opposite behaviour

About $\theta = 0$

$\sin \theta \approx \theta$:

$$ml\ddot{\theta} + kl\dot{\theta} + mg\theta = 0$$

Stable — the restoring term opposes any displacement.

About $\theta = \pi$

$\theta = \pi + \phi$, $\sin \theta \approx -\phi$:

$$ml\ddot{\phi} + kl\dot{\phi} - mg\phi = 0$$

Sign **flipped**: ϕ grows without bound — unstable.

Common pitfall

A linearization is only valid near the point it was taken about. Quoting a linearized model without stating its operating point is a meaningless answer.

Summary and what comes next

Summary — the framework and electrical/mechanical systems

- Every LTI causal model has the shape $a_n y^{(n)} + \dots + a_0 y = b_m u^{(m)} + \dots + b_0 u$; its characteristic equation depends only on the plant.
- Electrical networks: KVL/KCL with $v_R = Ri$, $v_L = L\dot{i}$, $i_C = C\dot{v}_C$. Translational and rotational mechanical systems: Newton's second law with $f = Kx$, $B\dot{x}$, $M\ddot{x}$ (or the rotational analogues).
- Force–voltage analogy: $M \leftrightarrow L$, $B \leftrightarrow R$, $K \leftrightarrow 1/C$ — same dynamics, different gain and units.

Summary — gears, the servomotor, and linearization

- Gear train ratio $n = \theta_2/\theta_1$: torque reflects by $1/n$, inertia and friction by n^2 .
- The d.c. servomotor couples an electrical equation (KVL with back e.m.f.) to a mechanical one (torque balance); eliminating the armature current gives the full transfer function, reducing to $K_m/[s(\tau_ms + 1)]$ when $L_a \rightarrow 0$.
- Linearization replaces a nonlinear equation by its first-order Taylor expansion about a stated operating point — the result, and even its stability, can differ completely between operating points.

Before the tutorial

You should be able to attempt these before you walk in (notes, “Before the tutorial”):

- 1 Derive the differential equation and transfer function $I(s)/V(s)$ of a *parallel RLC* network driven by a current source.
- 2 A rotational system has $J = 0.4 \text{ kg m}^2$, $b = 1.2 \text{ N m s rad}^{-1}$, no restoring spring. Find $\theta(s)/T(s)$ and its time constant.
- 3 A valve has $q = k\sqrt{p}$. Linearize about an operating point p_0 .

The tutorial itself works a full servomotor derivation (including the validity of $L_a \rightarrow 0$), a two-stage gear train, and two further linearization examples — a tank and a nonlinear-drag rotational load.

Looking ahead

- **Week 3** — these differential-equation models become transfer functions directly via the Laplace transform; the impedance shortcut for networks; poles, zeros, and the pole-zero map.
- **Assignment 1** (due start of Week 3 lecture) — Questions 3 and 4 use this week's material directly: network and mass-spring-damper modelling, and the geared d.c. servomotor.

The boundary, stated plainly

FEE3411 owns analysis; FEE3412 owns design. Deriving these models is analysis. Using a linearized model to design a compensator, or handling parameter uncertainty in the model itself, is FEE3412's business.

Reading for this week

Primary

Khalil, §2-1 (p. 10); §2-2.1 Electric Circuits (p. 14); §2-2.2 Mechanical Systems (p. 24); §2-2.3 Electromechanical Systems (p. 37); §2-4 Linearization (p. 50).

Secondary

Nise (6th ed.), §2.4 (p. 47); §2.5 (p. 61); §2.6 (p. 69); §2.7 Gears (p. 74); §2.8 Electromechanical (p. 79); §2.10–2.11 Linearization (pp. 88–89).

Further

Sundararajan, Ch. 3, Ch. 4, §6.4.